

Scalability & Future Plan

Date	7 July 2026
Team ID	xxxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-threaded local in-memory prediction execution.	Concurrent application loads from underwriting backlogs stall API worker routing threads.	High
2	No persistent database transaction storage.	Cardholder checks details and audit histories clear when the local server restarts.	High
3	Hard-coded business rules for delinquency checks.	Changing compliance policy thresholds requires server source code rebuilds.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask worker handling local requests.	Deploy Gunicorn workers on Render cloud with horizontal autoscaling configurations.
2	Data Storage	Ephemeral cache arrays and session arrays.	Integrate a relational PostgreSQL database to persist evaluation logs.
3	Performance	Synchronous thread-blocking scoring calls.	Configure asynchronous tasks (Celery + Redis) to queue multi-record batch checks.
4	Security	Standard HTTP REST routes without logins.	Implement JWT authentication modules, TLS encryption, and secure headers.

Future Roadmap:

To ensure continuous improvement, the project roadmap establishes a structured rollout plan across four consecutive development phases:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Watson ML Integration	Month 1	Migrate predictions from local pickle loads to cloud IBM Watson machine learning deployment endpoints, enhancing security.
Phase 2	Relational Database Setup	Month 3	Deploy persistent database ledgers to store historical evaluations and compliance logs forever.
Phase 3	Real-time Bureau API Sync	Month 6	Integrate active credit bureau API systems to replace manual customer input wizard validation.
Phase 4	Deep Learning Optimization	Month 12	Deploy deep neural networks models to predict custom loan line weights dynamically based on risk profiles.